Estimation

Reading Objectives:

In this reading, you will learn about the estimation of parameters based on sample outcomes. Estimation is of two types: point estimation and interval estimation. You will understand the derivations of these two estimations. You will also learn about interval estimation involving mean and proportion.

Main Reading Section:

1. Estimation: Introduction

Theory of estimation was founded by Prof R A Fisher.

Estimation of population parameters like μ (mean of the population) and σ^2 (variance of population) from the corresponding sample statistics has significant applications, which helps us to draw statistical inferences from the data given.

Statistical inference consists of those methods by which one makes inferences or generalizations about a population. The process of drawing inferences about a population on the basis of sample data is called statistical inference.

The two main classes of statistical inference are **estimation** of parameters and **testing hypotheses**.

Estimation can be either a **point estimator** that gives a single number estimate of the value of the parameter or an **interval estimate** that specifies an interval of plausible values for the parameter.

The sample mean $\overline{x}$ is an estimator of population mean μ.

The sample proportion p is an estimator of population proportion P = x/n.

The estimator of the population variance is $\frac{n}{n-1}S^2$, where S^2 is the sample variance.

A test of hypotheses provides the answer to whether the data support or contradict an investigator's claim about the value of the parameter.

2. Point Estimation: in which a single value is considered an estimate.

3. Interval Estimation—Mean: where an interval with a lower limit and an upper limit is considered as an estimate.

$$P\left(-z_{\alpha/2} \le \frac{\overline{X}-\mu}{\sigma/\sqrt{n}} \le z_{\alpha/2}\right) = 1-\alpha$$

rewriting the event as

$$\left[-z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \le \overline{X}-\mu \le z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right] = \left[\overline{X}-z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \le \mu \le \overline{X}+z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right]$$

we have

$$P\left(\overline{X}-z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \le \mu \le \overline{X}+z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right) = 1-\alpha$$

This last probability statement concerns a random interval covering the unknown parameter μ with probability $1-\alpha$.

Large sample confidence interval for μ (σ known)

$$\boxed{\overline{x}-z_{\alpha/2}\cdot\frac{\sigma}{\sqrt{n}} < \mu < \overline{x}+z_{\alpha/2}\cdot\frac{\sigma}{\sqrt{n}}}$$

Large sample confidence interval for μ

$$\bar{x} - z_{\alpha/2} \cdot \frac{s}{\sqrt{n}} < \mu < \bar{x} + z_{\alpha/2} \cdot \frac{s}{\sqrt{n}}$$

Small sample confidence interval for μ of normal population

$$\bar{x} - t_{\alpha/2} \cdot \frac{s}{\sqrt{n}} < \mu < \bar{x} + t_{\alpha/2} \cdot \frac{s}{\sqrt{n}}$$

Example: Consider a sample of size 50 with a mean of 305.58 and a variance of 1,366.86. Construct a 99% confidence interval or population mean.
Solution: n = 50, $\overline{x}$= 305.58, and s^2 = 1366.86

Z value from normal distribution tables at 99% = 2.575

Large sample confidence interval for μ

$$\bar{x} - z_{\alpha/2} \cdot \frac{s}{\sqrt{n}} < \mu < \bar{x} + z_{\alpha/2} \cdot \frac{s}{\sqrt{n}}$$

$$305.58 - 2.575.\frac{36.97}{\sqrt{50}} < \mu < 305.58 + 2.575.\frac{36.97}{\sqrt{50}}$$

$$292.12 < \mu < 319.04$$

Example: Consider a sample of size 18 with a mean of 22.6 and a standard deviation of 15.7. Construct a 95% confidence interval or population mean.
Solution: n = 18, $\overline{x}$= 22.6, and s = 15.7

t-value from distribution tables at 95% = 2.110 for n – 1 = 18 – 1 = 17 degrees of freedom

Small sample confidence interval for μ of normal population

$$\bar{x} - t_{\alpha/2} \cdot \frac{s}{\sqrt{n}} < \mu < \bar{x} + t_{\alpha/2} \cdot \frac{s}{\sqrt{n}}$$

$$22.6 - 2.110.\frac{15.7}{\sqrt{18}} < \mu < 22.6 + 2.110.\frac{15.7}{\sqrt{18}}$$

$$14,79 < \mu < 30.41$$

4. Interval Estimation: Proportion

When n is large, we can construct approximate confidence intervals for the binomial parameter p by using the normal approximation to the binomial distribution. Accordingly, we can assert that approximately

$$P\left(-z_{\alpha/2} < \frac{X - np}{\sqrt{np(1-p)}} < z_{\alpha/2}\right) = 1 - \alpha$$

Solving this quadratic inequality for p, we can obtain a corresponding set of approximate confidence limits for p in terms of the observed value x.

Large sample confidence interval for p

$$\frac{x}{n} - z_{\alpha/2}\sqrt{\frac{\frac{x}{n}\left(1-\frac{x}{n}\right)}{n}} < p < \frac{x}{n} + z_{\alpha/2}\sqrt{\frac{\frac{x}{n}\left(1-\frac{x}{n}\right)}{n}}$$

where the degree of confidence is $(1-\alpha)100\%$.

Example: If x = 36 of n = 100 persons interviewed are familiar with the tax exemptions given by the government for startups. Construct a 95% confidence interval for the corresponding true proportion.

Solution: x/n = 36 /100 = 0.36

Z value from the table is 1.96 at a 95% level of confidence.

Large sample confidence interval for p

$$\frac{x}{n} - z_{\alpha/2}\sqrt{\frac{\frac{x}{n}\left(1-\frac{x}{n}\right)}{n}} < p < \frac{x}{n} + z_{\alpha/2}\sqrt{\frac{\frac{x}{n}\left(1-\frac{x}{n}\right)}{n}}$$

where the degree of confidence is $(1-\alpha)100\%$.

$$0.36 - 1.96\sqrt{\frac{(0.36)(0.64)}{100}} < p < 0.36 + 1.96\sqrt{\frac{(0.36)(0.64)}{100}}$$

$$0.266 < p < 0.454$$